

Feel It, See It, Recover

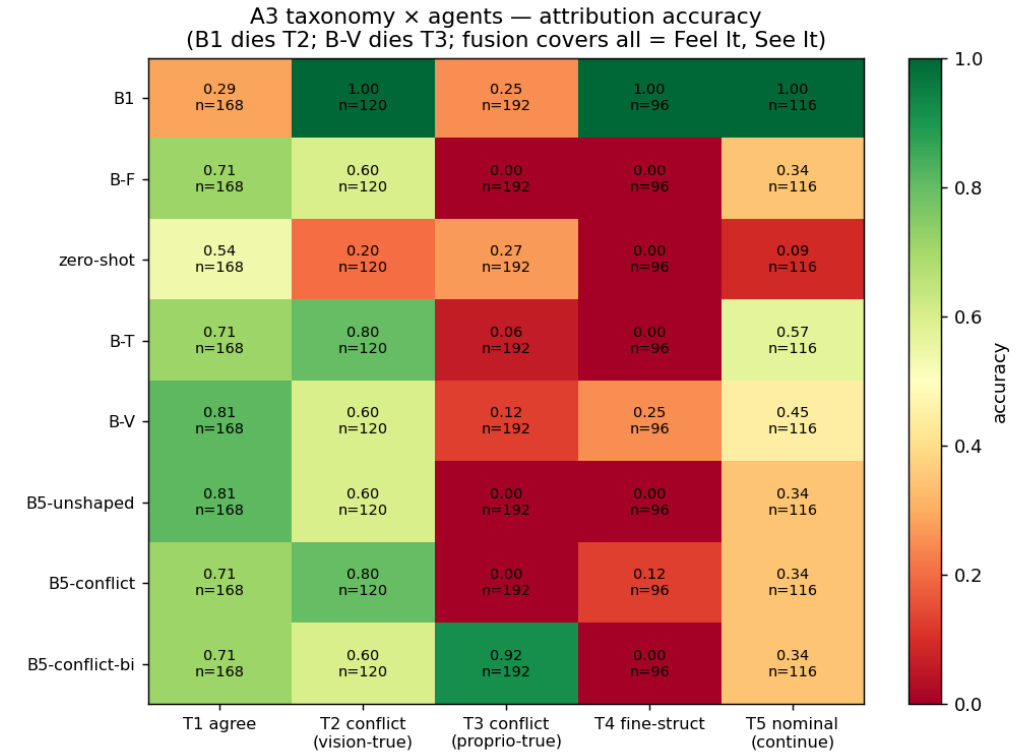
Cross-Modal Failure Attribution for Safe Quadrupedal-Navigation Recovery

- **The problem.** A legged robot walking over unknown terrain sometimes fails — it slips, gets stuck, or is overloaded. To recover **safely** it must first diagnose **why** — and the honest evidence is split across two senses that often **disagree**.
- **Two senses, each with blind spots.**
 - **Proprioception** — the robot's own body sensing (joint torques, foot slip, IMU) at **1000 Hz**: fast and always-on, but it **cannot see**.
 - **Vision (RGB-D camera)** — reads the surface look, but is fooled by **appearance**: a patch that looks safe but is slippery ice; a glass wall that is invisible.
- **This work's thesis.** Resolving these cross-modal conflicts — deciding which sense to trust, **per situation** — is a distinct cognitive operation that is:
 - (1) **necessary** — each sense has failures only the other can solve;
 - (2) **learned** — merely plugging in a camera is not enough;
 - (3) **consequential** — the diagnosis causally determines whether the robot falls / gets stuck / reaches the goal;
 - (4) **generalizes** and can self-assess its own uncertainty.
- **Contributions (this talk):** a certified benchmark + a learned cross-modal attributor (KinoVLA) + a measured safe-recovery guarantee — all on a physically-simulated Unitree Go2 with real RTX rendering and real CLIP vision.

Neither Sense Suffices

A bidirectional necessity result (Claim C1)

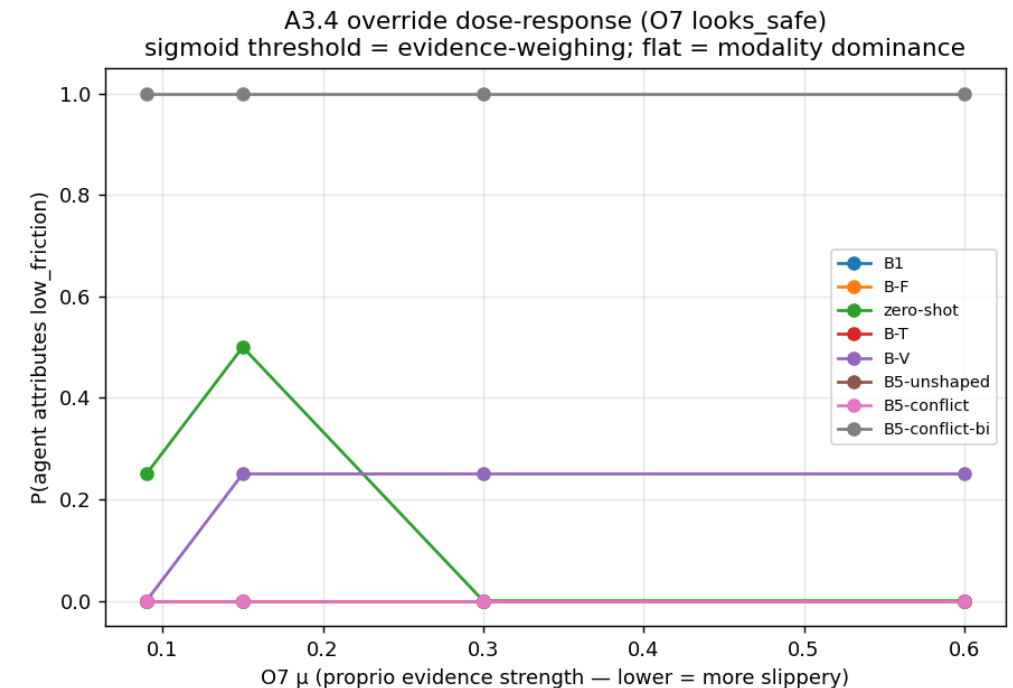
- **We organize every failure by which sense decides it (the *conflict taxonomy*):**
 - **T2 — vision-decidable:** the body-feel is *genuinely ambiguous* — two different hazards feel identical — only the camera separates them.
 - **T3 — body-decidable:** the camera is *deceived* (slippery ice disguised as safe ground; an invisible glass wall) — only the body-feel reveals it.
- **Constructive proof (the matched pair).** An adhesive / sticky surface (needs back-away) and soft mud (needs high-step-through) are tuned to exert the **exact same force** on the robot's legs.
 - $\Rightarrow$ the body-sensor signal is **byte-for-byte identical** (max difference = **0.0**); yet real CLIP vision separates them perfectly (AUC = **1.0**). Body-feel alone is provably blind.
- **Mirror proof (T3).** On a patch that *looks* like safe ground but is slippery ice, vision is blind (CLIP AUC = **0.31**) while body-feel is decisive (AUC = **1.0**); on an invisible wall, body-feel detects it (**100%**), vision cannot (**0%**).
- $\Rightarrow$ **Necessity is bidirectional:** no single modality subsumes the other. "Feel It AND See It."



Having Eyes Is Not Using Them

Conflict resolution is a learned skill (Claims C2, C5)

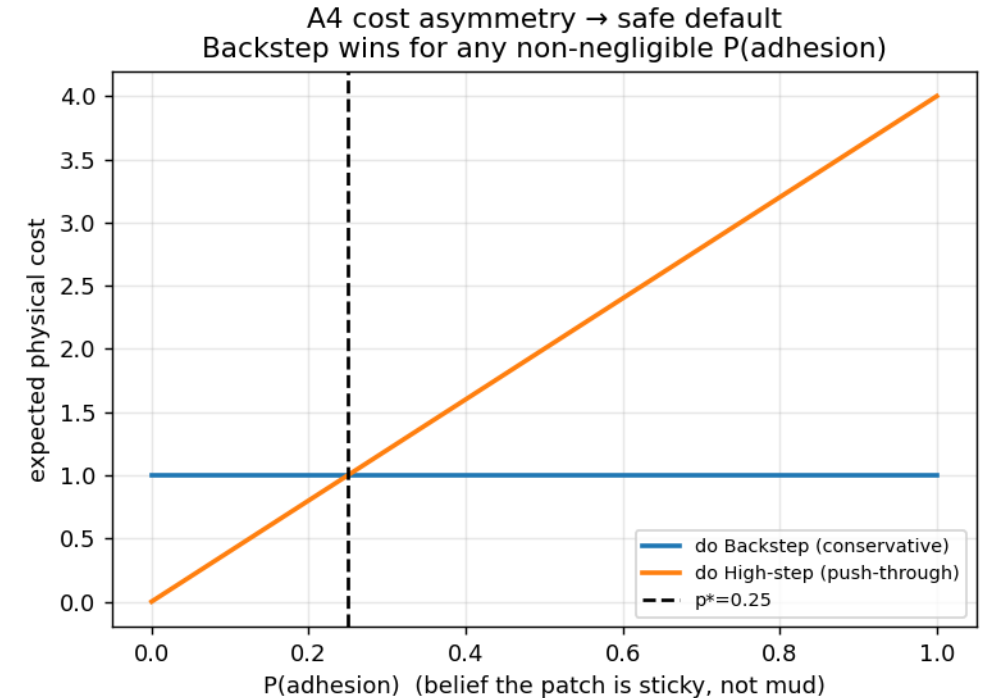
- **Claim:** adding a camera does **not** solve conflicts — the model must be **taught** to weigh evidence against appearance.
- **Three-row result** (balanced accuracy — a metric immune to gaming):
 - **Body-only agent: 0.50** — provably capped (on a byte-identical pair it can only output one constant guess).
 - **Four "has-vision" agents** (incl. a frozen CLIP+body fusion): all plateau at **0.80** — they solve the vision-decidable cases but crack nothing more.
 - **Our full model** (latent **Kino-Tokens** + conflict **chain-of-thought**): **0.90** — the only one that breaks the ceiling. Paired McNemar test $p < 1e-5$; reproduced across **3 seeds (0.900 / 0.900 / 0.900)**.
- **Shortcut-killer (appearance held-out):** trained on some surface looks, tested on **never-seen** looks → still **0.83** ⇒ it learned the **physics concept**, not a color cue. It even beats a frozen CLIP classifier on looks CLIP cannot separate ⇒ genuine reasoning.
- **Bidirectional evidence-weighting (figure):** a model trained only on "trust-vision" conflicts scores **0** on "trust-body" conflicts; trained **bidirectionally** it correctly **overrides a benign look when the body-signal is strong** (probability of "slippery" rises monotonically as slip grows). Genuine weighing — not "always trust the camera."



Wrong Diagnosis → Wrong Recovery

Attribution causally determines safety (Claim C3) + the intervention boundary (C4)

- **The label-swap matrix M.** Hold the physical scene fixed, **force** each recovery action (a *label*), measure the real outcome on the robot — isolating the causal effect of the diagnosis. **630 real-robot episodes**, deterministic (reproducible).
- **The diagonal dominates:** each hazard's *correct* recovery succeeds (rate **1.0**); wrong recoveries fail (**0.0**) — statistically separated.
- **The headline asymmetry:**
 - Pushing-through (correct for **mud**) on a **sticky** surface → the robot gets **stuck** (cost **4×**);
 - Backing-out (correct for **sticky**) on **mud** → merely **slow** (cost **1×**).
- ⇒ **mis-diagnosis cost is asymmetric.**
- **A safe default emerges (figure).** When unsure whether a patch is sticky or mud, **backing-out** is the expected-cost-minimizing action as soon as sticky-surface belief exceeds just **25%** — a decision-theoretic "Safe" rule derived from *measured consequences* (no hard-coded safety filter needed).
- **Intervention boundary:** the severity at which the robot's low-level adaptation stops being enough is a clean, agent-independent, reproducible number — $\theta^* = 0.275$.



Generalizes — and Knows When It Doesn't

Calibrated abstention + a strict safety advantage (Claim C4)

- **Generalization.** Attribution holds across **held-out surface appearances** (vision-dependent agents degrade most, body-only least — confirming body-feel is the appearance-invariant channel) and across **stacked operator combinations** (100% dominant-factor recognition).
- **A usable "I'm uncertain" signal.** A small privileged-physics predictor's prediction-error is a calibrated uncertainty score — it detects the model's own attribution errors with **AUROC 0.87**.
- **Abstain → safe default (figure).** When uncertain, fall back to back-away: the calibrated policy **strictly dominates** both *always-act-on-the-diagnosis* and *never-intervene*, on expected physical cost — turning the asymmetric mis-diagnosis cost into a **strict safety advantage**.
- **The system (real stack):** 1 kHz proprioception → **Kino-Tokens** (compressed body-signal tokens) → a **VLA recovery planner** (Qwen3-VL-4B + LoRA); evaluated on a physically-simulated Unitree Go2 + real RTX rendering + real CLIP; every number reproducible (deterministic reset).
- **One line: Feel It** (body-feel is necessary + appearance-invariant), **See It** (vision is necessary + appearance-decidable), **Recover** (the diagnosis causally determines safe recovery) — with a **calibrated escape hatch** when uncertain.

